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Guidelines for using GenAI tools in Learning and Assessment

AI in TLA Working Group

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GenAI and Large Language Models

Generative AI (GenAI): AI systems generating text, code, audio, images, videos

Large Language Models (LLMs): Main AI model used nowadays to generate language, code, and images.

- Generates text outputs in response to user prompts
- Trained on massive amounts of data from the internet
- Rely on language patterns to generate likely output.
- ChatGPT, Claude, Gemini

https://en.wikipedia.org/wiki/Generative_artificial_intelligence ; https://en.wikipedia.org/wiki/Large_language_model

Definitions and terms

- **Generative Artificial Intelligence (GenAI or AI):** System capable of generating text, images, videos, and other output by extrapolating from training data [2]
- **AI tools:** Software, libraries, packages or apps based on AI systems that deploy specific applications of some of these systems [2]
- **Plagiarism:** Use of ideas or material of others without acknowledgement
- **Self-plagiarism:** Re-use of one's own previously evaluated or published material without acknowledgement or indication thereof [1]
- **AI-giarism**: AI-enabled plagiarism [2]

Sources: [1] SU policy on plagiarism [2] SU position statement: Ethical Use of Artificial Intelligence

Background

- Rapid emergence of GenAI is reshaping higher education landscape
- Science requires analytical thinking, problem-solving, experimentation, and innovation
- GenAI and LLM technologies present both extraordinary opportunities and significant challenges in terms of ethical use, academic integrity, and pedagogical value.

SU policies and guidelines

- SU Policy on Plagiarism
- SU Position Statement on Ethical Use of AI in Research-Learning-Assessment
- Interim SU guidelines on allowable AI use and academic integrity

Background (cont'd)

- GenAI tools cannot and should not be prohibited
- Must learn to use these tools responsibly and critically
- Ethical use of GenAI tools should not be penalized
  - Penalize non-ethical and non-disclosed use
  - Excessive and non-critical use in assessments
- Considerations about what constitute permissible use of GenAI tools and what action to take when non-permissible use is suspected rest ultimately on SU's Plagiarism Policy

Principles for academic use of GenAI

LECTURER Guide AI-independent engagement where it is important for learning. Design critical thinking tasks.

STUDENT Make sure you learn and develop skills by engaging independently. Avoid AI over-reliance.

RESEARCH Reflect human understanding and view.

Authenticity

Promote human thinking. AI should support, not replace human engagement and learning.

LECTURER Encourage critical conversation and model responsible AI use.

Accountability

The user is reponsible for correctness, removing bias and avoiding plagiarism (by citing original sources)

STUDENT Take responsibility for work you create.

RESEARCH The authors are accountable. Use reproducible data/image processing.

Openness

Promote open use of AI systems and tools.

LECTURER Carefully consider how AI use might advantage or disadvantage students.

STUDENT Know the rules. Ensure that AI-use does not misrepresent your abilities or understanding.

Integrity and ethics

Don't use if prohibited or if it may be interpreted as falsification, misleading unfair or unethical.

RESEARCH Obey the policies of publishers. (e.g. AI modification of scientific data or images is not permitted. AI not permitted in peer-review.)

Transparency

Declare AI use. Guidelines on AI use must be communicated clearly.

LECTURER Clearly communicate guidelines (with reasons), Declare own AI use in teaching.

STUDENT Declare the use of AI tools and outputs and extent of use.

RESEARCH Declare AI use. Keep record of significant AI-use history.

When to declare AI use: Weak vs Strong use

- **Weak use**
  - Use of AI tools to assist in writing or correcting text or computer code
  - Tools not used to generate a substantial part of the work or code
  - Examples: Grammarly, Copilot
  - No need to declare, although declaration is recommended
- **Strong use**
  - Whole content (text, image, code, music) generated
  - User has little to no input in the creation process
  - Examples: Use of ChatGPT to generate large pieces of text, summaries or reviews
  - Must be declared, including purpose of use. Plagiarism if not declared.
  - Even if declared, extensive AI use may be regarded poor academic practice

Distinction between **weak** and **strong** not clear-cut. When in doubt, declare!

Declaration does not absolve students from responsibility of accuracy and validity of work.

Sources: [2] SU position statement: Ethical Use of Artificial Intelligence

Examples of AI use in academic writing

	Scenario	Plagiarism?	Why?
1	A student asks ChatGPT to write an essay or report and submits it unchanged or with limited changes.	Yes	GenAI content is not original work. Lack of transparency (AI use is not disclosed) and accountability (the work is not the student's own).
2	A student asks ChatGPT to write parts of an essay/report, rework these parts on their own, and submit it with a declaration at the end stating that AI was used in the development of the work.	No	GenAI content used but in the context of work authored by the student. AI use must be declared. Student must be able to explain which part is original and which part was influenced GenAI content.

Important:

- Amount of changes needed to qualify as original depends on the nature of the work and changes made.
- Should be considered in the context of SU's Plagiarism Policy. This is a general issue, not limited to GenAI.

Examples of AI use in academic writing

	Scenario	Plagiarism?	Why?
3	A student asks ChatGPT to summarise a journal article and submits the summary as their own.	Yes	This is still copying an AI-generated interpretation without citing it. The work submitted is not the student's own work.
4	A student uses ChatGPT for grammar and language suggestions.	No	Weak use of GenAI that need not be declared.
5	A student generates ideas with the help of GenAI for an essay/report; then writes the essay themselves, citing proper sources.	No	Using AI for planning and brainstorming is allowed. Good practice to declare use at the end of work submitted.

Examples of AI use in code generation and programming

	Scenario	Plagiarism?	Why?
6	A student copies AI-generated code into their project without attribution.	Yes	Just like text, code requires proper citation. AI-generated code is not original work. Lack of transparency (AI use is not disclosed) and accountability (the work submitted is not the student's own).
7	A student uses ChatGPT or copilot as part of a coding environment to develop a piece of code or program, prompting the AI for help and suggestions on specific parts of the code. They submit it with an AI use declaration.	No	Similar to Scenario 2: Code generated using AI then used into a larger code developed by the student. AI use must be declared. Student must be able to explain which part of the code or program is original and which part was influenced by AI-generated content.

Examples of AI use in code generation and programming

	Scenario	Plagiarism?	Why?
8	A student uses ChatGPT or copilot to debug a program.	**No**	Weak use of GenAI similar to a grammar tool that need not be declared.
9	A student uses ChatGPT or copilot obtain the first version of a program that is then refined with or without further help from the GenAI tool used.	**No**	Similar to planning or brainstorming. Use of AI permitted as long as the final code submitted has been reworked enough from the original AI-generated code and use of AI is declared.

Declare the use of AI tools

Always declare AI use and keep record of any significant AI-use history.

Be transparent

- Clearly state that you used AI tools.
- Place declaration at end of document

Describe how you used AI (see examples on next slide)

- Which tool(s) you used?
- What you used it for?
- How did you use, verify and modify the AI output?

Document your use

- Keep record of the prompts and outputs.

What not to do

- **Do not** submit AI-generated content without a declaration.
- **Do not** present AI-generated writing, code, or images as your own original work.
- **Do not** use AI tools in assessments where use is **explicitly prohibited** Examples: Take-home exams, lab reports, exam essays.

Examples of AI declarations

Grammar checking, sentence structuring, or vocabulary	***Minimal Assistance Disclosure*** *"I used GenAI (OpenAI's GPT-4) for proofreading and minor language refinement. All ideas, research, analysis, and substantive content remain entirely my own work."*
Brainstorming planning and idea generation	***Moderate Assistance Disclosure*** *"AI tools (identify tools used) were used to assist with brainstorming initial research questions and creating a preliminary outline. The final direction, research, analysis, and writing were conducted independently. Specific examples suggested by AI have been cited as 'Generated by Claude AI, Anthropic, 2024' where incorporated."*
Substantial content creation by AI. State how you checked, modified, revised, expanded it. (Make sure if this is acceptable and cite original sources to avoid plagiarism.)	***Substantial Assistance Disclosure*** *"Sections 2 and 4 were initially drafted with assistance from OpenAI's GPT-4. These sections were subsequently substantially revised, expanded, and edited to incorporate my own analysis and ensure alignment with my research findings. All statistical interpretations, conclusions, and key arguments represent my independent work and critical assessment of the data."*
Find, summarize and analyse sources, shape understanding o material. (Make sure if this is acceptable and cite original sources to avoid plagiarism.)	***Research Assistance Disclosure*** *"AI research tools were utilized to assist in summarizing and analyzing the 38 primary sources reviewed for this study. Each AI-generated summary was verified against the original source material, and all interpretations and conclusions drawn from these sources reflect my independent analysis rather than the AI's interpretation."*

Examples copied from *https://www.yomu.ai/resources/how-to-ethically-use-ai-essay-tools-while-maintaining-academic-integrity.*

Summary

- GenAI can help but misusing it leads to plagiarism.
- Line between permissible use and plagiarism not exact.
- Determined ultimately by being responsible and accountable for the work submitted, following SU's Plagiarism Policy.
- GenAI content must be declared.
- Be as open and specific as possible. The more open the better.
- GenAI should assist learning, not replace it.
- Always check AI-generated content: facts and references.

How to prompt an AI tool for useful output

Designing effective AI prompts is a skill

Prompt design

- Describe role GenAI should play (e.g. a lecturer, a tutor, an expert in, ...)
- Describe target audience
- Give as much detail about task as possible.
- Describe context or conditions.
- Give examples.
- Break down task into steps.
- Give format or structure of required output.

Iterative prompting

- Based on initial output, refine prompt.
- Ask for further clarification or detail.

Examples of AI prompting

Two examples of AI prompts for academic purposes

Using AI as a tutor

Example 1: I am a 4th-year student. Act as a tutor guiding me to solve this problem. Do not give me answers, but guide me by one question at a time. The problem is...

Example 2: I am a 4th-year student. Your role is a lecturer assessing my understanding of fluid mechanics. Ask me 5 conceptual questions, one at at time, to test my understanding on the concepts listed below. Do not give me the answers but guide me by questions. The concepts are...

Asking for feedback

Example 3: Act as an experienced peer reviewer for the journal ... and review this introduction. The aim of the introduction is to motivate why we investigated .... Evaluate the coherence, logical flow and clarity. Identify weaknesses of the discussion, explain which aspects can be improved, and what additional aspects could be addressed. Also provide detailed feedback on how the language and punctuation can be improved.

Note: *Read the output carefully. Do not accept automated rewriting by the AI tool. Apply the proposed corrections yourself so that you improve your writing skills in the process.*

How to evaluate and improve AI output

You are **accountable** for information from AI tools that you use. You must ensure **relevance**, **correctness, completeness** and **avoid bias**. You must find, read and **cite the original sources to avoid plagiarism**.

Relevance	• Is the output relevant to the task and the context? • What assumptions or approximations are made? • Are the assumptions relevant to the context?
Check correctness and avoid plagiarism	• Identify all facts that are not common knowledge. • Find reliable sources (official websites or publications) for these facts. • Read sources and check if they are relevant and support facts. • Add citations to sources. • Add deeper understanding obtained from reading sources.
Completeness	• Are there gaps? Are aspects mentioned, but not explained? • Are the assumptions or approximations made described?
Bias or discrimination	• Check for bias or discrimination and mitigate it.

When and how to cite original sources

Golden rule: If the idea or information has been published anywhere else then cite!

Direct quotes	Use quotation marks + cite the original source.	Example: Smith et al. [1] claims that "Abuse of GenAI is endangering science."
Paraphrased ideas	Cite the original source.	The unethical use of GenAI is considered a threat to scientific rigour [1]
Free online content	Follow the given attribution rules on how to cite.	
Code and software	Cite the authors, licenses, and repositories.	Based on code of J. Brown [2], modified for the purpose of...

References

[1] Smith et al., "Name of paper", Journal, vol x, xxx-xxx, year.

[2] J. Brown. Title of software, version, repository. Accessed: date. Available: URL

When and how to cite original sources

AI-generated text	Declare AI use AND cite the original sources. Usually we do **not** cite the AI tool.	See AI declarations examples.
	If you directly quote an AI output, or where AI output provided content for which no original source can be found, then cite the AI tool.	The discussion will analyse the following GenAI output [3]: Prompt: "What is the impact of..." Output: "The impact of recent advances..."

Reference example [3] OpenAI. *ChatGPT*. Version 4.5, OpenAI, 2025, https://chat.openai.com/

Exceptions where no citations are needed

Common knowledge. No citation needed.	"Liquid water is essential for the existence of life on earth."
Own novel idea. You cannot find any previous publications.	"According to my knowledge this is a novel concept and I will define it as..."

Bottom line

- Use every opportunity to **learn**.
- Understand, interpret, explain, analyze, question, and think critically.
- True **learning** is more than just finding **quick** answers.
- And that is why you are here!

Credits

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